

Gentle Floquet control of orbital Hall effect and orbital inverse Faraday effect

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Floquet engineering controls material responses via a time-periodic electromagnetic drive. An important task in practise is to reduce the driving strengths required to implement Floquet protocols. We show that in systems with strong phase-space constraints, where the response is dominated by a small region of momentum space, the relative impact of a resonant Floquet drive is enhanced. We demonstrate this idea on the example of the orbital Hall effect, where equilibrium occupations restrict the interband coherence to a narrow momentum interval $\hbar\Delta k$. The ratio $eE_0/(\Omega_0\hbar\Delta k)$, where Ω_0 is the frequency of the drive, is considerable even at modest field amplitudes E_0 , leading to a large relative change in the orbital Hall conductivity. Due to the non-trivial orbital texture, the circularly polarized drive induces an inverse orbital Faraday effect by magnetizing the electrons. We show that the corresponding susceptibility is controlled by the interband quantum metric that determines the resonant Floquet hybridization.

I. INTRODUCTION

Periodic driving with light is a versatile means for engineering electronic bands and inducing topological phases [1–8]. Floquet–Bloch states and light-induced Hall responses have been observed experimentally [9–14]. The preparation, occupations, and stability of driven states have motivated extensive work on pump protocols, relaxation, and nonequilibrium steady states [15–31]. Periodic driving has also been used to control correlated and symmetry-broken states [32–39].

The field amplitudes required for Floquet engineering depend on the electronic states and observables of interest. To modify electronic energies by an amount comparable to a typical bandwidth, the driving field must approach the microscopic electric fields acting on the electrons. Taking an energy of 1eV over a length of 1Å gives the estimate $E_0 \sim 10^{10}$ V/m [40]—an exceedingly high value. Weaker fields can be effective in manipulating low-energy degrees of freedom. Examples are Floquet topological insulators, where circularly polarized light is used to open topological bandgaps, that can be observed directly [9, 12–14] or through transport [11] at field amplitude of the order of $10^7 - 10^8$ V/m. Here the Floquet drive competes with broadening due to disorder and finite temperatures. Finally, the impact of Floquet drives can be enhanced by coupling to internal resonances [40–45].

We suggest that systems with strong phase-space constraints, whose transport properties are dominated by small, isolated “hot regions” of the Brillouin zone, provide one natural route around this limitation. In such systems, the drive need not substantially modify electronic states throughout the Brillouin zone, but only those within the restricted phase space controlling the response. To explore this idea, we study orbital Hall effects, which are often dominated by relatively small subregions of the Brillouin zone, either due to phase-space constraints or locally pronounced orbital Berry curvature [46], making them particularly susceptible to phase-space-selective Floquet engineering.

Orbitronics is emerging as a new direction in applied magnetism [47–49]. Building on concepts established in spintronics, but relying on orbital degrees of freedom, it promises fast, energy-efficient computation and memory that can be well integrated with existing technology [47]. Orbital angular momentum versions of well-known spintronic effects such as the orbital Hall effect [50, 51] and the orbital Edelstein effect [52, 53] do not necessarily require strong spin-orbit coupling and thus pose material constraints that are very different from their spin counterparts. They can, for example, be substantial even in light metals [46, 54, 55]. The idea of Floquet engineering the orbital Hall effect has been recently proposed [56]; however, we start from very different microscopic assumptions, and show under which circumstances the effects can be particularly large. We focus on the radial–tangential two-band Hamiltonian—a minimal model for momentum-dependent orbital textures and orbital Hall effects [57, 58]. Here, Pauli blocking confines the relevant interband processes to a small region in momentum space.

We denote the upper and lower band dispersions by $\varepsilon_+(k)$ and $\varepsilon_-(k)$ and their Fermi momenta by k_{F+} and k_{F-} , respectively. We assume that both bands have positive energies, such that the system has two nested Fermi surfaces (see the inset figure of Fig. 1). These two concentric Fermi surfaces enclose the “hot region” – an annulus of width

$$\Delta k = k_{F-} - k_{F+}, \quad (1)$$

where the lower band is occupied and the upper band is empty. The intrinsic, equilibrium interband response vanishes outside this annulus, as the bands are either both empty or both full, and the Pauli principle forbids interband transitions.

We consider driving with circularly polarized light of frequency Ω_0 and vector-potential amplitude $A_0 = E_0/\Omega_0$. The frequency defines a resonance momentum k_0 through

$$\varepsilon_+(k_0) - \varepsilon_-(k_0) = \Omega_0. \quad (2)$$

We will throughout consider the regime $k_{F+} < k_0 < k_{F-}$, which places the Floquet avoided crossing inside the annulus with nonzero interband response. It is convenient to define the momentum scale $q_A = eA_0/\hbar$ characterizing the driving amplitude. The central regime of this work is characterized by the two well-separated momentum scales Δk and k_0 , such that

$$q_A \sim \Delta k \ll k_0. \quad (3)$$

The condition $q_A \ll k_0$ ensures that the drive only weakly perturbs the underlying band structure. At the same time, it hybridizes states within a momentum region centered around k_0 , whose width is of order of q_A . Since the interband response of the non-driven system stems from states inside the annulus bounded by $k_{F\pm}$, the relative correction to the orbital Hall conductivity is of order $q_A/\Delta k$. Thus, a drive with $q_A \sim \Delta k$ yields an order-one relative change of the orbital Hall conductivity while remaining weak on the band scale.

II. INTRINSIC ORBITAL HALL EFFECT IN THE RADIAL-TANGENTIAL MODEL

In this section, we establish the orbital response of the undriven system, and identify the phase space that contributes to it. As a prototypical model valid in the vicinity of $\mathbf{k} = 0$, we consider the radial-tangential Hamiltonian [51, 57]. We show that Pauli blocking restricts the interband coherence responsible for the intrinsic orbital Hall effect to an annulus between two Fermi contours, whose width Δk is the momentum scale against which the amplitude of the Floquet drive will be compared.

The radial-tangential two-band model features two on-site orbitals with identical on-site energies, finite intra- and inter-orbital overlaps, and opposite effective angular momentum quantum numbers. An example would be two dipolar orbitals p_x and p_y with $m_\ell = \pm 1$ on a C_4 symmetric lattice. In the Bloch basis

$$|w_{x,y}(\mathbf{k})\rangle = \frac{1}{\sqrt{N}} \sum_n e^{i\mathbf{k}\cdot\mathbf{R}_n} |w_{x,y}(\mathbf{R}_n)\rangle, \quad (4)$$

where w_x and w_y are the two orbitals, the Hamiltonian reads [58]:

$$H_0(\mathbf{k}) = \begin{bmatrix} \frac{k_x^2 + k_y^2}{m_1} + \frac{k_x^2 - k_y^2}{m_2} & \frac{2k_x k_y}{m_2} \\ \frac{2k_x k_y}{m_2} & \frac{k_x^2 + k_y^2}{m_1} - \frac{k_x^2 - k_y^2}{m_2} \end{bmatrix}, \quad (5)$$

where $m_{1,2}$ are the effective masses determined by the orbital overlaps. The eigenvalues (band dispersions) are

$$\varepsilon_{\pm}(\mathbf{k}) = \left(\frac{1}{m_1} \pm \frac{1}{m_2} \right) k^2. \quad (6)$$

Here $k = |\mathbf{k}| = \sqrt{k_x^2 + k_y^2}$. We use units with $\hbar = 1$ from this point onward. Throughout this paper, we focus on

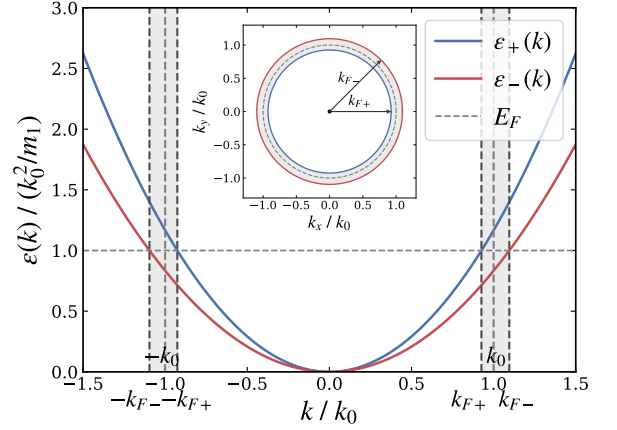


Figure 1. Band dispersions $\varepsilon_{\pm}(k)$ of the non-driven Hamiltonian Eq. (5) for $m_2 = 6m_1$. The horizontal dashed line marks E_F , and the vertical dashed lines mark the Fermi momenta defined in Eq. (7). The gray intervals, and the gray annulus in the inset, are the hot region $k_{F+} < k < k_{F-}$ in which $f_- - f_+ = 1$, cf. Eq. (8). The resonance momentum k_0 is chosen inside this annulus.

the regime $m_2 > m_1 > 0$, where both bands have positive curvature. For a positive Fermi energy $E_F > 0$, the model has two concentric, electron-like Fermi contours at

$$k_{F+} = \sqrt{\frac{E_F}{m_1^{-1} + m_2^{-1}}}, \quad k_{F-} = \sqrt{\frac{E_F}{m_1^{-1} - m_2^{-1}}}, \quad (7)$$

with $k_{F+} < k_{F-}$. We work at zero temperature with frozen equilibrium occupations. Consequently,

$$f_-(k) - f_+(k) = \begin{cases} 1, & k_{F+} < k < k_{F-}, \\ 0, & \text{otherwise.} \end{cases} \quad (8)$$

The band structure and Fermi contours are illustrated in Fig. 1.

The eigenvectors of $H_0(\mathbf{k})$, defining the band basis, are

$$\begin{aligned} |\psi_+(\mathbf{k})\rangle &= \cos \theta_k |w_x(\mathbf{k})\rangle + \sin \theta_k |w_y(\mathbf{k})\rangle, \\ |\psi_-(\mathbf{k})\rangle &= -i \sin \theta_k |w_x(\mathbf{k})\rangle + i \cos \theta_k |w_y(\mathbf{k})\rangle, \end{aligned} \quad (9)$$

where $k = |\mathbf{k}|$ and θ_k is the polar angle. In this basis, the Hamiltonian reads

$$H_0^{\text{band}}(\mathbf{k}) = (m_1^{-1} + m_2^{-1} \tau_3) k^2. \quad (10)$$

To compute transport, we employ the quantum Boltzmann equation (QBE) for the density matrix $\hat{\rho}$:

$$\frac{d\hat{\rho}}{dt} = -i[\hat{H}_0(t), \hat{\rho}] - i[\hat{H}_e, \hat{\rho}] + J[\hat{\rho}]. \quad (11)$$

Here $\hat{H}_e = e\mathbf{E}_{\text{dc}} \cdot \mathbf{r}$ accounts for the external DC electric field $\mathbf{E}_{\text{dc}}$. Without loss of generality, we assume that $\mathbf{E}_{\text{dc}} = \hat{e}_x E_{\text{dc}}$. The collision operator J describes scattering processes. Since we focus on the intrinsic interband

contribution to the orbital Hall response, we evaluate it in the collisionless limit. In the band basis, the collisionless QBE is [58, 59]:

$$\frac{d\rho^{\text{band}}(\mathbf{k}, t)}{dt} = -i[H_0^{\text{band}}(\mathbf{k}, t), \rho^{\text{band}}(\mathbf{k}, t)] + e\mathbf{E}_{\text{dc}} \cdot (\nabla_{\mathbf{k}} \rho^{\text{band}}(\mathbf{k}, t) - i[\mathcal{A}(\mathbf{k}), \rho^{\text{band}}(\mathbf{k}, t)]), \quad (12)$$

where the non-Abelian Berry connection is

$$\mathcal{A}_{ss'}(\mathbf{k}) = i\langle s\mathbf{k} | \nabla_{\mathbf{k}} | s'\mathbf{k} \rangle. \quad (13)$$

For the band basis in Eq. (9),

$$\mathcal{A}(\mathbf{k}) = \frac{\hat{e}_\theta}{k} \tau_1. \quad (14)$$

We denote the magnitude of its interband component by $\mathcal{A}_{+-}(k) \equiv |\mathcal{A}_{+-}(\mathbf{k})| = 1/k$.

Solving Eq. (12) with $d\rho/dt = 0$ yields the steady-state off-diagonal component [58]:

$$\rho_{+-}^{\text{band},(0)}(\mathbf{k}) = -\frac{\mathbf{E}_{\text{dc}} \cdot \mathcal{A}_{+-}(\mathbf{k})}{\varepsilon_+(\mathbf{k}) - \varepsilon_-(\mathbf{k})} (f_-(\mathbf{k}) - f_+(\mathbf{k})), \quad (15)$$

where $f_\pm$ are the equilibrium Fermi functions. By Eq. (8), $\rho_{+-}^{\text{band},(0)}$ is nonzero only in the annulus $k_{F+} < k < k_{F-}$. Thus the annulus of radial width $\Delta k = k_{F-} - k_{F+}$ contains the entire phase space contributing to the intrinsic interband response. This is the origin of the phase-space enhancement described in Eq. (3): a resonant region of width q_A modifies a fraction of order $q_A/\Delta k$ of the contributing states. In addition, the momentum integral reduces to

$$\int \frac{d^2\mathbf{k}}{(2\pi)^2} (f_- - f_+) (\dots) = \int_{k_{F+}}^{k_{F-}} \frac{k dk}{2\pi} \int_0^{2\pi} \frac{d\theta_k}{2\pi} (\dots). \quad (16)$$

In the band basis, the orbital magnetization and the longitudinal orbital currents flowing parallel to $\mathbf{E}_{\text{dc}}$ and orbital Hall currents flowing orthogonal to $\mathbf{E}_{\text{dc}}$ (see Fig. 2) are determined by the off-diagonal density matrix element (details see Appendix A):

$$\begin{aligned} m &= \mu_B \cdot \int \frac{d^2\mathbf{k}}{(2\pi)^2} 2\text{Re} \rho_{+-}^{\text{band}}(\mathbf{k}), \\ J_{Lz}^L &= \int \frac{d^2\mathbf{k}}{(2\pi)^2} \frac{4k_x}{m_1} \text{Re} \rho_{+-}^{\text{band}}(\mathbf{k}), \\ J_{Lz}^H &= \int \frac{d^2\mathbf{k}}{(2\pi)^2} \frac{4k_y}{m_1} \text{Re} \rho_{+-}^{\text{band}}(\mathbf{k}). \end{aligned} \quad (17)$$

Substitution of Eq. (15) into Eq. (17) gives

$$J_{Lz}^{L,(0)} = 0, \quad J_{Lz}^{H,(0)} = \frac{m_2 e E_{\text{dc}}}{2\pi m_1} \ln \frac{k_{F-}}{k_{F+}}. \quad (18)$$

The time-reversal-symmetric equilibrium state has zero orbital magnetization and no longitudinal orbital angular

momentum response. The intrinsic orbital Hall conductivity is

$$\sigma_{Lz}^{H,(0)} = \frac{J_{Lz}^{H,(0)}}{E_{\text{dc}}} = \frac{m_2 e}{2\pi m_1} \ln \frac{k_{F-}}{k_{F+}} = \frac{m_2 e}{4\pi m_1} \ln \frac{m_2 + m_1}{m_2 - m_1}. \quad (19)$$

For $m_1 \ll m_2$, $\sigma_{Lz}^{H,(0)}$ approaches $e/(2\pi)$. For a narrow annulus centered near k_0 , the logarithm scales as $\ln(k_{F-}/k_{F+}) \simeq \Delta k/k_0$. The undriven orbital Hall response is therefore proportional to the small phase-space fraction $\Delta k/k_0$, which makes its relative change particularly sensitive to a drive-induced momentum scale comparable to Δk .

III. RESONANT FLOQUET DRIVING OF THE RADIAL-TANGENTIAL MODEL

We now determine how the circularly polarized drive hybridizes the two bands near the one-photon resonance. We first derive the effective Floquet two-band Hamiltonian with its avoided level crossing, and then express the resonant coupling of the two bands in terms of the interband quantum metric. The resulting Floquet states provide the basis for calculating the orbital magnetization and orbital Hall response in the following sections.

We introduce an AC electric field characterized by the vector potential $\mathbf{A}(t)$. The Floquet Hamiltonian is obtained through Peierls substitution. We retain terms linear in A_0 in the Hamiltonian. This approximation is controlled by the smallness of eA_0/k_0 , even when eA_0 is comparable to Δk as in Eq. (3). In the band basis, the Hamiltonian reads

$$[H_0^{\text{band}}(\mathbf{k}, t)]_{ss'} = \langle \psi_s(\mathbf{k}) | \hat{H}_0(\mathbf{k} + e\mathbf{A}(t)) | \psi_{s'}(\mathbf{k}) \rangle. \quad (20)$$

To linear order in the driving field $\mathbf{A}$,

$$[H_0^{\text{band}}(\mathbf{k}, t)]_{ss'} = \delta_{ss'} \varepsilon_s(\mathbf{k}) + e\mathbf{A}(t) \cdot v_{ss'}(\mathbf{k}). \quad (21)$$

Here the velocity matrix element is

$$\begin{aligned} v_{ss'}(\mathbf{k}) &= \langle \psi_s(\mathbf{k}) | \nabla_{\mathbf{k}} \hat{H}_0(\mathbf{k}) | \psi_{s'}(\mathbf{k}) \rangle \\ &= \delta_{ss'} \nabla_{\mathbf{k}} \varepsilon_s(\mathbf{k}) + i(\varepsilon_s(\mathbf{k}) - \varepsilon_{s'}(\mathbf{k})) \mathcal{A}_{ss'}(\mathbf{k}). \end{aligned} \quad (22)$$

As shown in Fig. 2, we choose right-handed circular polarization, $\mathbf{A}(t) = A_0(\cos \Omega_0 t \hat{e}_x + \sin \Omega_0 t \hat{e}_y)$. Without loss of generality, $A_0 > 0$. The opposite helicity is obtained by changing the sign of the sine term, which reverses the induced orbital magnetization. The Floquet Hamiltonian for the chosen helicity becomes

$$\begin{aligned} [H_0^{\text{band}}(\mathbf{k}, t)]_{ss'} &= \delta_{ss'} \varepsilon_s(k) \\ &+ eA_0 \cos(\Omega_0 t - \theta_k) \delta_{ss'} \frac{d\varepsilon_s(k)}{dk} \\ &+ ieA_0 \sin(\Omega_0 t - \theta_k) (\varepsilon_s(k) - \varepsilon_{s'}(k)) \mathcal{A}_{ss'}(k). \end{aligned} \quad (23)$$

The Floquet spectrum is derived in Appendix B. The one-photon resonance occurs at

$$k_0 = \sqrt{m_2 \Omega_0 / 2}, \quad (24)$$

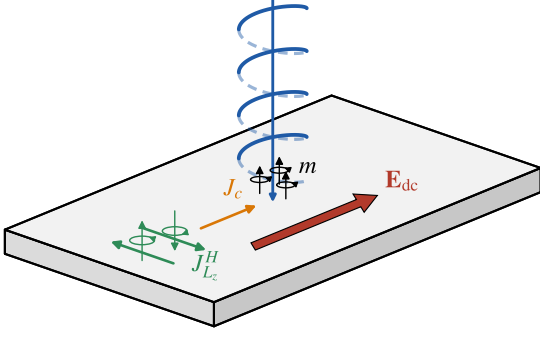


Figure 2. The suggested set-up. The blue helix represents the right-handed, circularly polarized AC field, the red arrow represents the DC probe field $\mathbf{E}_{dc}$, and the orange arrow represents the longitudinal charge flow. The circular drive induces a static out-of-plane orbital magnetization m – a phenomenon known as the inverse Faraday effect [60] – and modifies the orbital Hall current $J_{L_z}^H$. Reversing the optical helicity reverses m .

where $\varepsilon_+(k_0) - \varepsilon_-(k_0) = 2k_0^2/m_2 = \Omega_0$. We choose Ω_0 such that $k_{F+} < k_0 < k_{F-}$, placing the resonance in the annulus where the interband response is nonzero. The resulting quasienergies are

$$\begin{aligned} E_+(k) &= \bar{\varepsilon}(k) + \frac{\Omega_0 + \Delta(k)}{2}, \\ E_-(k) &= \bar{\varepsilon}(k) - \frac{\Omega_0 + \Delta(k)}{2}. \end{aligned} \quad (25)$$

Here, $\Delta(k)$ describes avoided crossing induced by the drive:

$$\Delta(k) = \text{sgn}(k - k_0) [(\Omega_0 - \Delta_0(k))^2 + (eA_0\Delta_0(k)\mathcal{A}_{+-}(k))^2]^{1/2}, \quad (26)$$

Here $\bar{\varepsilon}(k) = \frac{\varepsilon_+(k) + \varepsilon_-(k)}{2}$ is the band average dispersion, and $\Delta_0(k) = \varepsilon_+(k) - \varepsilon_-(k)$ is the static band gap. The sign in Eq. (26) follows from our choice of Floquet gauge and branch labels. We choose the Floquet representatives on the two sides of the resonance such that $E_{\pm}$ and $\Psi_{\pm}$ connect adiabatically to the undriven bands as $A_0 \rightarrow 0$. This requires interchanging the two resonant eigenvectors and shifting one quasienergy by Ω_0 across k_0 , as shown explicitly in Appendix B, Eqs. (B16) and (B17). The signed quantity $\Delta(k)$ keeps these band labels continuous through the resonance, while $|\Delta(k)|$ is the magnitude of the avoided-crossing gap. Substituting Eqs. (6) and (14) gives

$$\Delta(k) = \text{sgn}(k - k_0) \sqrt{\left(\Omega_0 - \frac{2k^2}{m_2}\right)^2 + \left(\frac{2keA_0}{m_2}\right)^2}. \quad (27)$$

The corresponding Floquet dispersion is shown in Fig. 3.

The drive-induced interband coupling can be expressed in terms of the quantum geometric tensor (QGT). For a

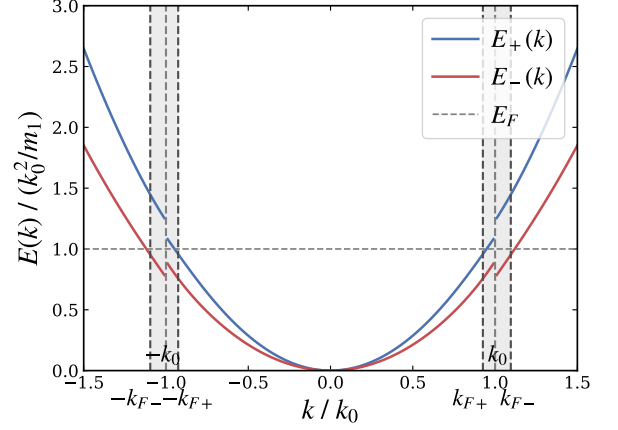


Figure 3. Floquet quasienergy bands $E_{\pm}(k)$ according to Eq. (25). The vertical dashed lines indicate k_{F+} , k_0 , and k_{F-} . The avoided crossing at k_0 lies inside the gray annulus $k_{F+} < k < k_{F-}$, where the interband response is nonzero. All parameters are the same as in Fig. 1.

two-band model, the QGT of band s reads [61]

$$Q_{ij}^{(s)} = \langle \partial_{k_i} \psi_s | (1 - |\psi_s\rangle\langle\psi_s|) | \partial_{k_j} \psi_s \rangle = \mathcal{A}_{s\bar{s},i} \mathcal{A}_{\bar{s}s,j}. \quad (28)$$

Its real part is the Fubini-Study quantum metric,

$$g_{ij}^{(s)} = \text{Re} Q_{ij}^{(s)}. \quad (29)$$

For a two-band model, the two bands have the same quantum metric, $g_{ij}^{(+)} = g_{ij}^{(-)} = g_{ij}$, while

$$g_{xx} + g_{yy} = \mathcal{A}_{+-,x} \mathcal{A}_{-+,x} + \mathcal{A}_{+-,y} \mathcal{A}_{-+,y} = \mathcal{A}_{+-}^2(k). \quad (30)$$

Therefore, the Floquet gap can be written as

$$\Delta(k) = \text{sgn}(k - k_0) [(\Omega_0 - \Delta_0(k))^2 + (eA_0\Delta_0(k))^2 \text{Tr}g(k)]^{1/2}. \quad (31)$$

Here $\text{Tr}g = g_{xx} + g_{yy}$.

IV. INVERSE ORBITAL FARADAY EFFECT

Circularly polarized light breaks time-reversal symmetry and induces a static orbital magnetization m , associated with the inverse orbital Faraday effect [62, 63] (see Fig. 2). We focus on the atomic contribution to the inverse Faraday magnetization, disregarding itinerant effects [64]. The expectation value of $\hat{L}_z$, time-averaged over one cycle of the drive, is given by

$$\langle \Psi_{\pm}^\dagger(\mathbf{k}) \hat{L}_z \Psi_{\pm}(\mathbf{k}) \rangle_{\text{cycle}} = \pm \frac{eA_0}{2\Omega_0} \frac{d\Delta_0(k)}{dk} \sin \phi(k), \quad (32)$$

where $\Psi_{\pm}(\mathbf{k})$ are the Floquet eigenstates defined in Eq. (B22), and the mixing angle $\phi(k)$ is defined as

$$\phi(k) = \arctan \frac{eA_0\Delta_0(k)\mathcal{A}_{+-}(k)}{\Omega_0 - \Delta_0(k)}, \quad \phi \in [-\pi/2, \pi/2]. \quad (33)$$

The derivation is given in Appendix C. Substituting Eqs. (6) and (14) gives

$$\begin{aligned} \langle \Psi_{\pm}^{\dagger}(\mathbf{k}) \hat{L}_z \Psi_{\pm}(\mathbf{k}) \rangle_{\text{cycle}} &= \pm \frac{2keA_0}{m_2\Omega_0} \\ &\times \frac{keA_0}{\sqrt{(keA_0)^2 + (k_0^2 - k^2)^2}} \text{sgn}(k - k_0). \end{aligned} \quad (34)$$

Since the two bands have opposite L_z expectation values, their contributions cancel wherever both are occupied or both are empty. The net magnetization therefore comes only from the annulus defined in Eq. (8) and shown in Fig. 1. For the right-handed polarization used above, the static orbital magnetization is thus

$$\begin{aligned} m &= \mu_B \int_{k_{F+}}^{k_{F-}} \frac{k dk}{2\pi} \int_0^{2\pi} \frac{d\theta_k}{2\pi} \times \left(-\frac{2keA_0}{m_2\Omega_0} \right) \\ &\times \frac{keA_0}{\sqrt{(keA_0)^2 + (k_0^2 - k^2)^2}} \text{sgn}(k - k_0), \end{aligned} \quad (35)$$

where μ_B is the Bohr magneton. The integral in Eq. (35) is finite throughout the closed interval $k_{F+} \leq k_0 \leq k_{F-}$. When the resonance is well separated from both Fermi contours, i.e.

$$\min(k_0 - k_{F+}, k_{F-} - k_0) \gg eA_0, \quad (36)$$

it has the asymptotic form derived in Appendix C:

$$m = \frac{\mu_B e^2 A_0^2}{2\pi m_2 \Omega_0} \left[k_0^2 \ln \frac{k_0^2 - k_{F+}^2}{k_{F-}^2 - k_0^2} - k_{F-}^2 + k_{F+}^2 \right] + O(A_0^3). \quad (37)$$

Writing the optical helicity as $\xi = \pm 1$, the result for both polarizations is

$$m = \xi \chi_{\text{IFE}} E_0^2 + O(E_0^3), \quad (38)$$

where the effective inverse Faraday susceptibility χ_{IFE} is

$$\chi_{\text{IFE}} = \frac{e^2 \mu_B}{4\pi \Omega_0^2} \left(\ln \frac{k_0^2 - k_{F+}^2}{k_{F-}^2 - k_0^2} - \frac{k_{F-}^2 - k_{F+}^2}{k_0^2} \right). \quad (39)$$

Here we used $\Omega_0 = 2k_0^2/m_2$ and $E_0 = \Omega_0 A_0$. The magnetization is quadratic in E_0 and reverses sign with the optical helicity. The logarithmic divergence as k_0 approaches either Fermi contour marks the breakdown of the asymptotic expansion Eq. (36). The exact integral in Eq. (35) remains finite.

We parameterize the resonance position within the annulus by

$$k_0 = \eta k_{F+} + (1 - \eta) k_{F-}, \quad 0 < \eta < 1. \quad (40)$$

Fig. 4 shows χ_{IFE} as a function of η . It decreases with η and changes sign at $\eta \simeq 0.410$. The exact finite-drive result approaches the asymptotic curve away from the Fermi contours and remains finite at both endpoints.

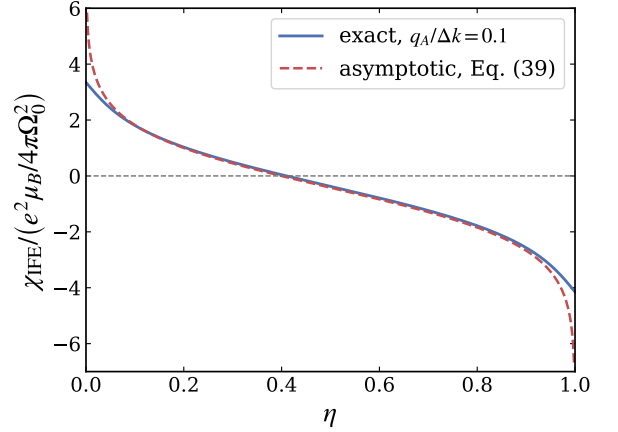


Figure 4. Inverse orbital Faraday susceptibility χ_{IFE} in units of $\frac{e^2 \mu_B}{4\pi \Omega_0^2}$ as a function of the resonance position η defined in Eq. (40). The solid curve is obtained by integration of Eq. (35) over the full interval at $q_A/\Delta k = 0.1$. The dashed curve is the asymptotic result of Eq. (39). We use the same m_1 , m_2 , and E_F as in Fig. 1.

V. FLOQUET CONTROL OF THE ORBITAL HALL EFFECT

We now calculate the orbital Hall conductivity in the presence of the AC drive. To obtain the response linear in the DC probe field E_{dc} , we substitute the time-dependent Hamiltonian $H_0^{\text{band}}(\mathbf{k}, t)$ given in Eq. (23) into the QBE Eq. (12), and retain terms linear in E_{dc} . We use the frozen equilibrium occupations in Eq. (8), thereby disregarding heating and the redistribution of occupations between Floquet bands. The full derivation is presented in Appendix D. Averaging over one driving period gives the off-diagonal density-matrix element entering the observables of Eq. (17):

$$\begin{aligned} \text{Re} \langle \rho_{+-}^{\text{band}}(\mathbf{k}) \rangle_{\text{cycle}} &= \frac{keA_0}{m_2\Omega_0} \sin \phi (n_- - n_+) \\ &- \cos \theta_k e E_{\text{dc}} \frac{keA_0 \sin \phi}{m_2\Omega_0} \tau_{f,0} \partial_k (n_- - n_+) \\ &+ \sin \theta_k e E_{\text{dc}} \frac{(n_- - n_+)}{k} \left(\frac{\Omega_0 \cos \phi - \Delta(1 + \cos^2 \phi)/2}{\Omega_0^2 - \Delta^2} \right. \\ &\quad \left. - \frac{keA_0 \sin \phi \cos \phi}{m_2\Omega_0 \Delta} \right). \end{aligned} \quad (41)$$

Here, $\tau_{f,0}$ is a phenomenological relaxation time, which we introduce to regularize the longitudinal response. We stress that the intrinsic orbital Hall response remains finite in the clean limit $\tau_{f,0} \rightarrow \infty$. The first term in Eq. (41) is present at $E_{\text{dc}} = 0$ and is independent of θ_k . It corresponds to the static orbital magnetization discussed in Sec. IV. Setting $n_- - n_+ = 1$ for $k_{F+} < k < k_{F-}$ and substituting into Eq. (17) gives exactly Eq. (35). The quantum Boltzmann and Floquet-state calculations

of the static magnetization therefore agree.

The second term in Eq. (41) is proportional to $\cos \theta_k$ and produces the longitudinal orbital angular momentum current. The third is proportional to $\sin \theta_k$ and produces the orbital Hall current. Because each term is multiplied by $n_- - n_+$ or its derivative, its radial integral is confined either to the annulus $k_{F+} < k < k_{F-}$ or to the two Fermi surfaces. Substituting Eq. (41) into Eq. (17) gives the conductivities (Appendix E):

$$\begin{aligned}\sigma_{Lz}^L &= \frac{J_{Lz}^L}{E_{\text{dc}}} = -\frac{e^3 \tau_{f,0} A_0^2}{\pi m_1 m_2 \Omega_0} [G_L(k_{F+}) + G_L(k_{F-})], \\ \sigma_{Lz}^H &= \frac{J_{Lz}^H}{E_{\text{dc}}} = \frac{m_2 e}{2\pi m_1} [2G_H(k_0) - G_H(k_{F-}) - G_H(k_{F+})].\end{aligned}\quad (42)$$

Here,

$$G_L(k) = \frac{k^4}{\sqrt{(k_0^2 - k^2)^2 + (eA_0 k)^2}}, \quad (43)$$

and $G_H(k)$ is defined in Eq. (E6). The longitudinal orbital conductivity is proportional to $A_0^2 \tau_{f,0}$ and describes the light-induced orbital polarization carried by the longitudinal flow.

The orbital Hall response depends on the position of the interband resonance. For $k_{F+} < k_0 < k_{F-}$, the Floquet resonance inside the integration interval produces a correction linear in the drive amplitude A_0 . If k_0 lies outside the annulus $k_{F+} < k < k_{F-}$, the integrand is smooth over the full integration range and the weak-drive correction starts at A_0^2 . We focus on the first situation. Expanding to leading order gives (see Appendix E)

$$\sigma_{Lz}^H \approx \frac{m_2 e}{2\pi m_1} \left(\ln \frac{k_{F-}}{k_{F+}} - \frac{3eA_0}{2k_0} \right), \quad (44)$$

where the linear-in- A_0 correction originates from the Floquet resonance inside the annulus of width Δk . For a narrow annulus centered near k_0 , $\ln(k_{F-}/k_{F+}) \simeq \Delta k/k_0$. Restoring $\hbar$ and recalling that

$$q_A = eA_0/\hbar, \quad (45)$$

Eq. (44) becomes

$$\frac{\sigma_{Lz}^H}{\sigma_{Lz}^{H,(0)}} \simeq 1 - \frac{3q_A}{2\Delta k}. \quad (46)$$

Thus $q_A \sim \Delta k$ produces an order-one relative change while $q_A/k_0 \ll 1$. The amplitude of an electric field corresponding to $q_A/\Delta k$ is

$$E_0 = \Omega_0 A_0 = \frac{\hbar \Omega_0}{e} \frac{q_A}{\Delta k} \Delta k. \quad (47)$$

The dependence of σ_{Lz}^H on A_0 is shown in Fig. 5. For $m_2 = 6m_1$, the curves with $\eta = 0.60, 0.70$, and 0.80 cross zero at $q_A/\Delta k \simeq 3.42, 2.67$, and 2.33 , respectively. The zero crossings lie beyond the weak-drive regime of

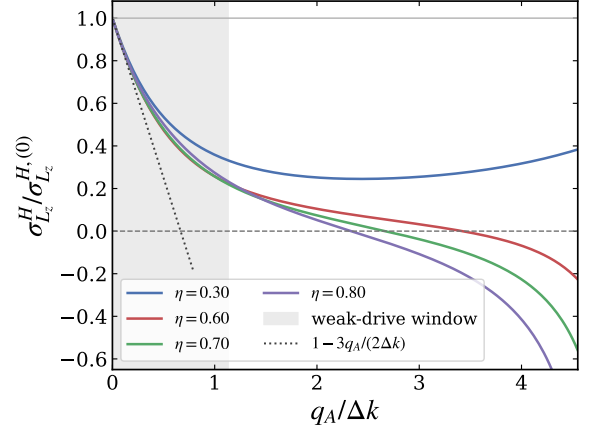


Figure 5. Orbital Hall conductivity in the presence of a Floquet drive, cf. Eq. (42), as a function of $q_A/\Delta k = eA_0/(\hbar\Delta k)$, normalized by its static value. η , defined in Eq. (40), describes the position of the resonance. The bandstructure parameters are the same as in Fig. 4. For $\hbar\Omega_0 = 0.2\text{eV}$ and $\Delta k = 0.01\text{\AA}^{-1}$, one unit on the horizontal axis corresponds to $E_0 = 2.0 \times 10^7\text{V/m}$ by Eq. (47). The gray region marks the interval in which $q_A/k_0 \leq 0.2$ for all plotted values of k_0 . The dotted line shows the approximated result of Eq. (46). The full result of Eq. (42) predicts that the conductivity changes sign depending on the resonance positions at stronger driving.

Eq. (44), and the plotted curves use the full expression in Eq. (42). Taking a representative mid-infrared photon energy $\hbar\Omega_0 = 0.2\text{eV}$ and a narrow-annulus width $\Delta k = 0.01\text{\AA}^{-1}$, Eq. (47) gives $E_0 \simeq 6.8 \times 10^7, 5.3 \times 10^7$, and $4.7 \times 10^7\text{V/m}$ for these three zero crossings. These fields are comparable to the $4.0 \times 10^7\text{V/m}$ peak field used with 191meV photons in Ref. [11]. This indicates that while large relative changes in the orbital Hall conductivity can be induced by relatively weak fields, the sign of the Hall conductivity can be reversed at larger, but still realistic field strengths.

VI. CONCLUSION AND OUTLOOK

Using the radial-tangential model, we have shown that resonant Floquet driving can efficiently modify the orbital Hall conductivity when Pauli blocking confines interband coherence to a narrow region of momentum space. The circularly polarized drive also generates an orbital magnetization whose weak-drive susceptibility is controlled by the quantum geometry of the system.

The present theory could be extended in a number of directions. First, it would be worthwhile to include effects of disorder, finite temperature, heating, and scattering, which will to varying degrees be present in actual experimental realisations. More broadly, the enhancement of Floquet effects via phase-space constraints can also arise in materials with more complex Fermi surfaces if Pauli blocking confines an interband-coherent response

to a narrow momentum region. In more complex settings, additional bands may introduce further resonances and modify the orbital texture and quantum geometry. Furthermore, hot regions with pronounced orbital Berry curvature could lead to similar enhancements. These effects will be considered in future work.

Finally, the strong dependence of the orbital Hall conductivity on the intensity of circularly polarized light is of interest for applications in orbitronics. It may, e.g., enable all-optical orbital-current switches and valves that could control and direct orbital angular momentum currents by locally modifying orbital Hall conductivities with light.

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Appendix A: Orbital angular momentum and current operators

The angular momentum operator $\hat{L}_z$ satisfies

$$\hat{L}_z |w_x(\mathbf{k})\rangle = i |w_y(\mathbf{k})\rangle, \quad \hat{L}_z |w_y(\mathbf{k})\rangle = -i |w_x(\mathbf{k})\rangle. \quad (\text{A1})$$

Therefore,

$$\begin{aligned} \hat{L}_z |s = +, \mathbf{k}\rangle &= i \cos \theta_k |w_y(\mathbf{k})\rangle - i \sin \theta_k |w_x(\mathbf{k})\rangle = |s = -, \mathbf{k}\rangle, \\ \hat{L}_z |s = -, \mathbf{k}\rangle &= \sin \theta_k |w_y(\mathbf{k})\rangle + \cos \theta_k |w_x(\mathbf{k})\rangle = |s = +, \mathbf{k}\rangle. \end{aligned} \quad (\text{A2})$$

Its matrix representation in the band basis is therefore

$$L_z^{\text{band}} = \tau_1. \quad (\text{A3})$$

Consequently, the orbital magnetization obtained from the density matrix is

$$m = \mu_B \int \frac{d^2 \mathbf{k}}{(2\pi)^2} \text{Tr} [L_z^{\text{band}} \rho^{\text{band}}(\mathbf{k})] = 2\mu_B \int \frac{d^2 \mathbf{k}}{(2\pi)^2} \text{Re} \rho_{+-}^{\text{band}}(\mathbf{k}). \quad (\text{A4})$$

For an internal observable O , the corresponding current matrix is

$$\mathbf{J}_O(\mathbf{k}) = \frac{1}{2} \{O, \nabla_{\mathbf{k}} H_0(\mathbf{k})\}, \quad (\text{A5})$$

and its expectation value is

$$\langle \mathbf{J}_O(\mathbf{k}) \rangle = \text{Tr}[\rho(\mathbf{k}) \mathbf{J}_O(\mathbf{k})]. \quad (\text{A6})$$

Therefore,

$$\begin{aligned} J_{L_z}^L &= \int \frac{d^2 \mathbf{k}}{(2\pi)^2} \frac{1}{2} \text{Tr} [(L_z \partial_{k_x} H_0(\mathbf{k}) + \partial_{k_x} H_0(\mathbf{k}) L_z) \rho(\mathbf{k})], \\ J_{L_z}^H &= \int \frac{d^2 \mathbf{k}}{(2\pi)^2} \frac{1}{2} \text{Tr} [(L_z \partial_{k_y} H_0(\mathbf{k}) + \partial_{k_y} H_0(\mathbf{k}) L_z) \rho(\mathbf{k})]. \end{aligned} \quad (\text{A7})$$

Using $H_0^{\text{band}}(\mathbf{k}) = (m_1^{-1} + m_2^{-1} \tau_3) k^2$ gives

$$J_{L_z}^L = \int \frac{d^2 \mathbf{k}}{(2\pi)^2} \frac{4k_x}{m_1} \text{Re} \rho_{+-}^{\text{band}}(\mathbf{k}), \quad J_{L_z}^H = \int \frac{d^2 \mathbf{k}}{(2\pi)^2} \frac{4k_y}{m_1} \text{Re} \rho_{+-}^{\text{band}}(\mathbf{k}). \quad (\text{A8})$$

Appendix B: Derivation of the effective Hamiltonian

The Floquet Hamiltonian is

$$[H_0^{\text{band}}(\mathbf{k}, t)]_{ss'} = \delta_{ss'} \varepsilon_s(k) + eA_0 \cos(\Omega_0 t - \theta_k) \cdot \delta_{ss'} \frac{d\varepsilon_s(k)}{dk} + ieA_0 \sin(\Omega_0 t - \theta_k) (\varepsilon_s(k) - \varepsilon_{s'}(k)) \mathcal{A}_{ss'}(k). \quad (\text{B1})$$

Its Fourier expansion is

$$H_0^{\text{band}}(\mathbf{k}, t) = h_0(\mathbf{k}) + \sum_{n=1}^{\infty} (h_n(\mathbf{k})e^{in\Omega_0 t} + h_n^\dagger(\mathbf{k})e^{-in\Omega_0 t}), \quad (\text{B2})$$

with

$$[h_0(\mathbf{k})]_{ss'} = \delta_{ss'}\varepsilon_s(k), \quad [h_1(\mathbf{k})]_{ss'} = \frac{1}{2}eA_0e^{-i\theta_k} \left(\delta_{ss'} \frac{d\varepsilon_s(k)}{dk} + (\varepsilon_s(k) - \varepsilon_{s'}(k))\mathcal{A}_{ss'}(k) \right), \quad h_{n>1}(\mathbf{k}) = 0. \quad (\text{B3})$$

The solution to the Floquet Schrödinger equation

$$i \frac{\partial \Psi(\mathbf{k}, t)}{\partial t} = H_0^{\text{band}}(\mathbf{k}, t) \Psi(\mathbf{k}, t) \quad (\text{B4})$$

is labeled by a Floquet eigenvalue E and a set of Fourier coefficients ψ_n :

$$\Psi(\mathbf{k}, t) = e^{-iE(\mathbf{k})t} \sum_{n=-\infty}^{\infty} \psi_n(\mathbf{k}) e^{in\Omega_0 t}. \quad (\text{B5})$$

The Fourier components $\psi_n(\mathbf{k})$ satisfy

$$\begin{bmatrix} \ddots & & & & & & \\ & h_0 + 2\Omega_0 & h_1 & 0 & 0 & 0 & \\ & h_1^\dagger & h_0 + \Omega_0 & h_1 & 0 & 0 & \\ & 0 & h_1^\dagger & h_0 & h_1 & 0 & \\ & 0 & 0 & h_1^\dagger & h_0 - \Omega_0 & h_1 & \\ & 0 & 0 & 0 & h_1^\dagger & h_0 - 2\Omega_0 & \\ & & & & & & \ddots \end{bmatrix} \begin{bmatrix} \vdots \\ \psi_2 \\ \psi_1 \\ \psi_0 \\ \psi_{-1} \\ \psi_{-2} \\ \vdots \end{bmatrix} = E \begin{bmatrix} \vdots \\ \psi_2 \\ \psi_1 \\ \psi_0 \\ \psi_{-1} \\ \psi_{-2} \\ \vdots \end{bmatrix}. \quad (\text{B6})$$

Suppressing the momentum arguments and using the real interband connection in Eq. (9) gives

$$\begin{bmatrix} \ddots & & & & & & \\ & \bar{\varepsilon} - \frac{\Delta_0}{2} + 2\Omega_0 & -e^{-i\theta_k}V_A & e^{-i\theta_k}V_- & 0 & 0 & 0 \\ & -e^{i\theta_k}V_A & \bar{\varepsilon} + \frac{\Delta_0}{2} + \Omega_0 & 0 & e^{-i\theta_k}V_+ & e^{-i\theta_k}V_A & 0 \\ & e^{i\theta_k}V_- & 0 & \bar{\varepsilon} - \frac{\Delta_0}{2} + \Omega_0 & -e^{i\theta_k}V_A & e^{-i\theta_k}V_- & 0 \\ & 0 & e^{i\theta_k}V_+ & -e^{i\theta_k}V_A & \bar{\varepsilon} + \frac{\Delta_0}{2} & 0 & e^{-i\theta_k}V_+ \\ & 0 & e^{i\theta_k}V_A & e^{i\theta_k}V_- & 0 & \bar{\varepsilon} - \frac{\Delta_0}{2} & -e^{-i\theta_k}V_A \\ & 0 & 0 & 0 & e^{i\theta_k}V_+ & -e^{i\theta_k}V_A & \bar{\varepsilon} + \frac{\Delta_0}{2} - \Omega_0 \\ & & & & & & \ddots \end{bmatrix} \begin{bmatrix} \vdots \\ \psi_{2,-} \\ \psi_{1,+} \\ \psi_{1,-} \\ \psi_{0,+} \\ \psi_{0,-} \\ \psi_{-1,+} \\ \vdots \end{bmatrix} = E \begin{bmatrix} \vdots \\ \psi_{2,-} \\ \psi_{1,+} \\ \psi_{1,-} \\ \psi_{0,+} \\ \psi_{0,-} \\ \psi_{-1,+} \\ \vdots \end{bmatrix}. \quad (\text{B7})$$

Here $\bar{\varepsilon}(k) = [\varepsilon_+(k) + \varepsilon_-(k)]/2$, $\Delta_0(k) = \varepsilon_+(k) - \varepsilon_-(k)$, $V_\pm = (eA_0/2)d\varepsilon_\pm(k)/dk$, and $V_A = eA_0\Delta_0(k)\mathcal{A}_{+-}(k)/2$.

Near resonance, where $\Delta_0 \approx \Omega_0$, the central 2×2 block of Eq. (B7) is

$$\begin{bmatrix} \bar{\varepsilon} - \frac{\Delta_0}{2} + \Omega_0 & -e^{-i\theta_k}V_A \\ -e^{i\theta_k}V_A & \bar{\varepsilon} + \frac{\Delta_0}{2} \end{bmatrix} \begin{bmatrix} \psi_{1,-} \\ \psi_{0,+} \end{bmatrix} = E \begin{bmatrix} \psi_{1,-} \\ \psi_{0,+} \end{bmatrix}. \quad (\text{B8})$$

The solutions are

$$\begin{aligned} E'_+ &= \bar{\varepsilon} + \frac{\Omega_0}{2} + \sqrt{\frac{(\Omega_0 - \Delta_0)^2}{4} + V_A^2}, & \begin{bmatrix} \psi_{1,-} \\ \psi_{0,+} \end{bmatrix}_+ &= \begin{bmatrix} \cos(\phi'/2) \\ -\sin(\phi'/2)e^{i\theta_k} \end{bmatrix}, \\ E'_- &= \bar{\varepsilon} + \frac{\Omega_0}{2} - \sqrt{\frac{(\Omega_0 - \Delta_0)^2}{4} + V_A^2}, & \begin{bmatrix} \psi_{1,-} \\ \psi_{0,+} \end{bmatrix}_- &= \begin{bmatrix} \sin(\phi'/2) \\ \cos(\phi'/2)e^{i\theta_k} \end{bmatrix}. \end{aligned} \quad (\text{B9})$$

Here the mixing angle is

$$\phi' = \arctan \frac{2V_A}{\Omega_0 - \Delta_0}, \quad \phi' \in [0, \pi]. \quad (\text{B10})$$

The central subspace $\{|-, 1\rangle, |+, 0\rangle\}$ couples through the intraband matrix elements to $|-, 2\rangle$ and $|+, -1\rangle$. These two couplings are represented by

$$\begin{bmatrix} e^{-i\theta_k} V_- & 0 \\ 0 & e^{-i\theta_k} V_+ \end{bmatrix}. \quad (\text{B11})$$

The interband matrix element also couples the central subspace to $|+, 2\rangle$ and $|-, -1\rangle$, represented by

$$\begin{bmatrix} 0 & 0 \\ e^{-i\theta_k} V_A & 0 \end{bmatrix}. \quad (\text{B12})$$

These four states are separated from the central subspace by energies of order Ω_0 or $2\Omega_0$. We therefore evaluate the energy denominators at $A_0 = 0$. Their drive-induced corrections enter at higher order. To first order in A_0 , perturbation theory gives

$$\begin{aligned} \psi_{2,+} &= -\frac{e^{-i\theta_k} V_A}{2\Omega_0} \psi_{1,-}, & \psi_{2,-} &= -\frac{e^{-i\theta_k} V_-}{\Omega_0} \psi_{1,-}, & \psi_{1,+} &= -\frac{e^{-i\theta_k} V_+}{\Omega_0} \psi_{0,+}, \\ \psi_{0,-} &= \frac{e^{i\theta_k} V_-}{\Omega_0} \psi_{1,-}, & \psi_{-1,+} &= \frac{e^{i\theta_k} V_+}{\Omega_0} \psi_{0,+}, & \psi_{-1,-} &= \frac{e^{i\theta_k} V_A}{2\Omega_0} \psi_{0,+}. \end{aligned} \quad (\text{B13})$$

Using the definition Eq. (B5) of the Floquet eigenstate, we combine the amplitudes in Eqs. (B9) and (B13) and obtain

$$\Psi'_+ = e^{-iE'_+ t} \begin{bmatrix} -\sin(\phi'/2) e^{i\theta_k} (1 + iK_+ + L_2'^*) \\ \cos(\phi'/2) e^{i\Omega_0 t} (1 + iK_- - L_1'^*) \end{bmatrix}, \quad \Psi'_- = e^{-iE'_- t} \begin{bmatrix} \cos(\phi'/2) e^{i\theta_k} (1 + iK_+ - L_1') \\ \sin(\phi'/2) e^{i\Omega_0 t} (1 + iK_- + L_2') \end{bmatrix}. \quad (\text{B14})$$

Here

$$K_\pm = \frac{2V_\pm}{\Omega_0} \sin(\theta_k - \Omega_0 t), \quad L_1' = \frac{V_A}{2\Omega_0} \tan \frac{\phi'}{2} e^{-2i(\theta_k - \Omega_0 t)}, \quad L_2' = \frac{V_A}{2\Omega_0} \cot \frac{\phi'}{2} e^{2i(\theta_k - \Omega_0 t)}. \quad (\text{B15})$$

As $A_0 \rightarrow 0$, the eigenvalues $E'_\pm$ defined in Eq. (B9) approach Floquet-zone-shifted branches. Floquet periodicity identifies E and $E + n\Omega_0$ ($n \in \mathbb{Z}$) as equivalent eigenvalues, with eigenstates ψ and $e^{-in\Omega_0 t}\psi$, respectively. Define k_0 as the resonance momentum where $\Omega_0 = \Delta_0(k_0)$. We fix the Floquet gauge and branch labels by requiring $E_\pm$ and $\Psi_\pm$ to connect adiabatically to the undriven upper and lower bands, respectively, as $A_0 \rightarrow 0$. Because the two resonant branches exchange their undriven-band character at k_0 , this convention selects different representatives of the raw Floquet solutions for $k < k_0$ and $k > k_0$. The resulting eigenvalues are

$$E_+ = \begin{cases} \bar{\varepsilon} + \frac{\Omega_0}{2} - \sqrt{\frac{(\Omega_0 - \Delta_0)^2}{4} + V_A^2}, & k < k_0; \\ \bar{\varepsilon} + \frac{\Omega_0}{2} + \sqrt{\frac{(\Omega_0 - \Delta_0)^2}{4} + V_A^2}, & k > k_0. \end{cases} \quad E_- = \begin{cases} \bar{\varepsilon} - \frac{\Omega_0}{2} + \sqrt{\frac{(\Omega_0 - \Delta_0)^2}{4} + V_A^2}, & k < k_0; \\ \bar{\varepsilon} - \frac{\Omega_0}{2} - \sqrt{\frac{(\Omega_0 - \Delta_0)^2}{4} + V_A^2}, & k > k_0. \end{cases} \quad (\text{B16})$$

The corresponding Floquet-gauge representatives are

$$\Psi_+ = \begin{cases} \Psi'_-, & k < k_0; \\ \Psi'_+, & k > k_0. \end{cases} \quad \Psi_- = \begin{cases} \Psi'_+ e^{-i\Omega_0 t}, & k < k_0; \\ \Psi'_- e^{-i\Omega_0 t}, & k > k_0. \end{cases} \quad (\text{B17})$$

The sign in the definition

$$\Delta(k) = \text{sgn}(k - k_0) \cdot \sqrt{(\Omega_0 - \Delta_0)^2 + 4V_A^2} = \text{sgn}(k - k_0) \sqrt{(\Omega_0 - \Delta_0)^2 + (eA_0\Delta_0(k)\mathcal{A}_{+-}(k))^2}, \quad (\text{B18})$$

therefore records the exchange of Floquet representatives across the resonance. The branch convention in Eq. (B17) makes $\Delta(k)$ a signed quantity. The physical gap is $|\Delta(k)|$. This definition combines the $k < k_0$ and $k > k_0$ branches of Eq. (B16) into

$$E_\pm(k) = \bar{\varepsilon}(k) \pm \frac{\Omega_0 + \Delta(k)}{2}. \quad (\text{B19})$$

Similarly, the redefinition

$$\phi = \arctan \frac{2V_A}{\Omega_0 - \Delta_0}, \quad \phi \in [-\frac{\pi}{2}, \frac{\pi}{2}] \quad (\text{B20})$$

together with

$$L_1 = \frac{V_A}{2\Omega_0} \tan \frac{\phi}{2} e^{-2i(\theta_k - \Omega_0 t)}, \quad L_2 = \frac{V_A}{2\Omega_0} \cot \frac{\phi}{2} e^{2i(\theta_k - \Omega_0 t)} \quad (\text{B21})$$

yields

$$\Psi_+ = e^{-iE_+ t} \begin{bmatrix} -\sin(\phi/2)e^{i\theta_k}(1 + iK_+ + L_2^*) \\ \cos(\phi/2)e^{i\Omega_0 t}(1 + iK_- - L_1^*) \end{bmatrix}, \quad \Psi_- = e^{-iE_- t} \begin{bmatrix} \cos(\phi/2)e^{i\theta_k}(1 + iK_+ - L_1) \\ \sin(\phi/2)e^{i\Omega_0 t}(1 + iK_- + L_2) \end{bmatrix}. \quad (\text{B22})$$

Appendix C: Cycle-averaged orbital angular momentum and static magnetization

The expectation values of $\hat{L}_z$ in the Floquet states of Eq. (B22) are

$$\begin{aligned} & \Psi_\pm^\dagger \hat{L}_z \Psi_\pm \\ &= \mp \sin \frac{\phi}{2} \cos \frac{\phi}{2} \left(e^{i(\theta_k - \Omega_0 t)}(1 + iK_+ + L_2^*)(1 - iK_- - L_1) + e^{-i(\theta_k - \Omega_0 t)}(1 - iK_+ - L_1^*)(1 + iK_- + L_2) \right) \\ &= \mp \sin \frac{\phi}{2} \cos \frac{\phi}{2} \left(e^{i(\theta_k - \Omega_0 t)}(1 + iK_+ - iK_- - L_1 + L_2^*) + e^{-i(\theta_k - \Omega_0 t)}(1 - iK_+ + iK_- - L_1^* + L_2) \right) + O(A_0^2) \\ &= \pm \left((K_+ - K_-) \sin \phi \sin(\theta_k - \Omega_0 t) - \sin \phi \cos(\theta_k - \Omega_0 t) + \sin \phi \operatorname{Re}((L_1 - L_2^*)e^{i(\theta_k - \Omega_0 t)}) \right) + O(A_0^2). \end{aligned} \quad (\text{C1})$$

Using Eq. (B15),

$$\Psi_\pm^\dagger \hat{L}_z \Psi_\pm = \pm \left(2 \sin \phi \frac{V_+ - V_-}{\Omega_0} \sin^2(\theta_k - \Omega_0 t) - \left(\sin \phi + \frac{V_A \cos \phi}{\Omega_0} \right) \cos(\theta_k - \Omega_0 t) \right). \quad (\text{C2})$$

Averaging over one driving period gives

$$\langle \Psi_\pm^\dagger \hat{L}_z \Psi_\pm \rangle_{\text{cycle}} = \pm \frac{V_+ - V_-}{\Omega_0} \sin \phi. \quad (\text{C3})$$

Using the definitions in Appendix B gives

$$\langle \Psi_\pm^\dagger(\mathbf{k}) \hat{L}_z \Psi_\pm(\mathbf{k}) \rangle_{\text{cycle}} = \pm \frac{eA_0}{2\Omega_0} \frac{d\Delta_0(k)}{dk} \frac{eA_0\Delta_0(k)\mathcal{A}_{+-}(k)}{\sqrt{(eA_0\Delta_0(k)\mathcal{A}_{+-}(k))^2 + (\Omega_0 - \Delta_0(k))^2}} \operatorname{sgn}(k - k_0). \quad (\text{C4})$$

The magnetization is

$$\begin{aligned} m &= \mu_B \int_{k_{F+}}^{k_{F-}} \frac{k dk}{2\pi} \int_0^{2\pi} \frac{d\theta_k}{2\pi} \left(-\frac{2keA_0}{m_2\Omega_0} \right) \frac{keA_0}{\sqrt{(keA_0)^2 + (k_0^2 - k^2)^2}} \operatorname{sgn}(k - k_0) \\ &= \frac{\mu_B e A_0}{\pi m_2 \Omega_0} \int_{k_{F+}}^{k_{F-}} \frac{k^3 e A_0}{\sqrt{(keA_0)^2 + (k_0^2 - k^2)^2}} \operatorname{sgn}(k_0 - k) dk \\ &= \frac{\mu_B e^2 A_0^2}{\pi m_2 \Omega_0} \cdot \frac{1}{2} (F(k_0) - F(k_{F+}) - F(k_{F-}) + F(k_0)) \\ &= \frac{\mu_B e^2 A_0^2}{2\pi m_2 \Omega_0} (2F(k_0) - F(k_{F+}) - F(k_{F-})). \end{aligned} \quad (\text{C5})$$

Here,

$$\begin{aligned} F(k) &= \int \frac{2k^3}{\sqrt{(keA_0)^2 + (k_0^2 - k^2)^2}} dk \\ &= \sqrt{(k^2 - k_0^2)^2 + (eA_0 k)^2} + \left(k_0^2 - \frac{(eA_0)^2}{2} \right) \ln \left(2\sqrt{(k^2 - k_0^2)^2 + (eA_0 k)^2} + 2(k^2 - k_0^2) + (eA_0)^2 \right). \end{aligned} \quad (\text{C6})$$

Therefore

$$\begin{aligned}
F(k_0) &= eA_0k_0 + k_0^2 \ln(2eA_0k_0 + (eA_0)^2) + O(A_0^2), \\
F(k_{F+}) &= k_0^2 - k_{F+}^2 + k_0^2 \ln \left(2(k_{F+}^2 - k_0^2) + 2 \left(k_0^2 - k_{F+}^2 + \frac{(eA_0k_{F+})^2}{2(k_0^2 - k_{F+}^2)} \right) + (eA_0)^2 \right) + O(A_0^2) \\
&= k_0^2 - k_{F+}^2 + k_0^2 \ln \frac{(eA_0k_0)^2}{k_0^2 - k_{F+}^2} + O(A_0^2), \\
F(k_{F-}) &= k_{F-}^2 - k_0^2 + k_0^2 \ln(4(k_{F-}^2 - k_0^2)) + O(A_0^2),
\end{aligned} \tag{C7}$$

and

$$\begin{aligned}
2F(k_0) - F(k_{F+}) - F(k_{F-}) &= k_0^2 \ln \frac{(2eA_0k_0)^2}{4(k_{F-}^2 - k_0^2) \cdot \frac{(eA_0k_0)^2}{k_0^2 - k_{F+}^2}} - k_{F-}^2 + k_{F+}^2 + 2eA_0k_0 + O(A_0^2) \\
&= k_0^2 \ln \frac{k_0^2 - k_{F+}^2}{k_{F-}^2 - k_0^2} - k_{F-}^2 + k_{F+}^2 + 2eA_0k_0 + O(A_0^2).
\end{aligned} \tag{C8}$$

Substituting into Eq. (C5) gives Eq. (37).

Appendix D: Solution of the Floquet QBE

For the AC-driven system, the Hamiltonian in Eq. (12) is the periodic $H_0^{\text{band}}(\mathbf{k}, t)$ in Eq. (20). We solve this Floquet QBE in the periodic modes

$$|\Phi_{\pm}(\mathbf{k}, t)\rangle = e^{iE_{\pm}(\mathbf{k})t} |\Psi_{\pm}(\mathbf{k}, t)\rangle. \tag{D1}$$

They satisfy $(i\partial_t + E_m(\mathbf{k}))|\Phi_m(\mathbf{k}, t)\rangle = H_0^{\text{band}}(\mathbf{k}, t)|\Phi_m(\mathbf{k}, t)\rangle$. Eq. (B22) gives

$$|\Phi_m(\mathbf{k}, t)\rangle = \sum_{s=\pm} V_{sm}(\mathbf{k}, t) |\psi_s(\mathbf{k})\rangle, \quad \mathbf{V}(\mathbf{k}, t) = \begin{bmatrix} \cos(\phi/2)e^{i\theta_k}(1 + iK_+ - L_1) & -\sin(\phi/2)e^{i\theta_k}(1 + iK_+ + L_2^*) \\ \sin(\phi/2)e^{i\Omega_0 t}(1 + iK_- + L_2) & \cos(\phi/2)e^{i\Omega_0 t}(1 + iK_- - L_1^*) \end{bmatrix}. \tag{D2}$$

Here

$$\begin{aligned}
\phi(k) &= \arctan \frac{2keA_0}{m_2\Omega_0 - 2k^2}, \quad \phi \in [-\frac{\pi}{2}, \frac{\pi}{2}], \quad K_{\pm} = 2 \frac{(m_1^{-1} \pm m_2^{-1})keA_0}{\Omega_0} \sin(\theta_k - \Omega_0 t), \\
L_1 &= \frac{keA_0}{2m_2\Omega_0} \tan(\phi/2)e^{-2i(\theta_k - \Omega_0 t)}, \quad L_2 = \frac{keA_0}{2m_2\Omega_0} \cot(\phi/2)e^{2i(\theta_k - \Omega_0 t)}.
\end{aligned} \tag{D3}$$

In this basis, $\rho_{mm'}^F(\mathbf{k}, t) = \langle \Phi_m(\mathbf{k}, t) | \hat{\rho} | \Phi_{m'}(\mathbf{k}, t) \rangle$, and the Floquet QBE becomes

$$\frac{d\rho_{mm'}^F(\mathbf{k}, t)}{dt} + i(E_m(\mathbf{k}) - E_{m'}(\mathbf{k}))\rho_{mm'}^F(\mathbf{k}, t) = e\mathbf{E}_{\text{dc}} \cdot (\nabla_{\mathbf{k}} \rho_{mm'}^F(\mathbf{k}, t) - i[\mathcal{A}^F(\mathbf{k}, t), \rho^F(\mathbf{k}, t)]_{mm'}). \tag{D4}$$

The Floquet Berry connection is

$$\mathcal{A}_{mm'}^F(\mathbf{k}, t) = \sum_{ss'} V_{ms}^\dagger(\mathbf{k}, t) \mathcal{A}_{ss'}(\mathbf{k}) V_{s'm'}(\mathbf{k}, t) + i \sum_s V_{ms}^\dagger(\mathbf{k}, t) \nabla_{\mathbf{k}} V_{sm'}(\mathbf{k}, t) = \mathbf{V}^\dagger \mathcal{A} \mathbf{V} + i \mathbf{V}^\dagger \nabla_{\mathbf{k}} \mathbf{V}. \tag{D5}$$

The first term corresponds to a unitary transformation of the static Berry connection, and the second term arises from the $\mathbf{k}$ -dependence of the transformation matrix itself.

Substitution of Eq. (D2) gives

$$\mathcal{A}^F(\mathbf{k}, t) = \sum_{M=-\infty}^{\infty} \begin{bmatrix} \mathbf{X}_M(k) & \mathbf{Y}_M(k) \\ \mathbf{Y}_{-M}^*(k) & -\mathbf{X}_M(k) \end{bmatrix} \cdot e^{iM(\theta_k - \Omega_0 t)}. \tag{D6}$$

The components entering the linear response are

$$\begin{aligned}
\mathbf{Y}_1 &= -i \frac{eA_0 \sin \phi}{m_2 \Omega_0} \hat{e}_k + \left[-\sin^2 \frac{\phi}{2} + \frac{ekA_0 \sin \phi}{2m_2 \Omega_0} \right] \frac{\hat{e}_\theta}{k} + O(A_0^2), \\
\mathbf{Y}_{-1} &= i \frac{eA_0 \sin \phi}{m_2 \Omega_0} \hat{e}_k + \left[\cos^2 \frac{\phi}{2} + \frac{ekA_0 \sin \phi}{2m_2 \Omega_0} \right] \frac{\hat{e}_\theta}{k} + O(A_0^2), \\
\mathbf{Y}_0 &= -\frac{1}{2} i \frac{\partial \phi}{\partial k} \hat{e}_k + \frac{1}{2} \sin \phi \frac{\hat{e}_\theta}{k} + O(A_0), \\
\mathbf{Y}_{M \neq 0, \pm 1} &= O(A_0).
\end{aligned} \tag{D7}$$

The density matrix is expanded as

$$\rho^F(\mathbf{k}, t) = \sum_{M=-\infty}^{\infty} \begin{bmatrix} F_M(\mathbf{k}) + f_M(\mathbf{k}) & g_M(\mathbf{k}) \\ g_{-M}^*(\mathbf{k}) & F_M(\mathbf{k}) - f_M(\mathbf{k}) \end{bmatrix} \cdot e^{iM(\theta_k - \Omega_0 t)}. \tag{D8}$$

Substituting these two equations into the Floquet QBE Eq. (D4) yields a set of coupled algebraic equations for the functions $F_M(\mathbf{k})$, $f_M(\mathbf{k})$, and $g_M(\mathbf{k})$:

$$\begin{aligned}
(-\tau_{F,M}^{-1} - iM\Omega_0)F_M &= eE_{dc} \left[\cos \theta_k \frac{\partial F_M}{\partial k} - \frac{\sin \theta_k}{k} \left(\frac{\partial F_M}{\partial \theta_k} + iMF_M \right) \right], \\
(-\tau_{f,M}^{-1} - iM\Omega_0)f_M &= eE_{dc} \left[\cos \theta_k \frac{\partial f_M}{\partial k} - \frac{\sin \theta_k}{k} \left(\frac{\partial f_M}{\partial \theta_k} + iMf_M \right) \right] - i \sum_{M'=-\infty}^{\infty} e\mathbf{E}_{dc} \cdot (g_{-(M-M')}^* \mathbf{Y}_{M'} - g_{M-M'} \mathbf{Y}_{-M'}^*), \\
i(\Delta - M\Omega_0)g_M &= eE_{dc} \left[\cos \theta_k \frac{\partial g_M}{\partial k} - \frac{\sin \theta_k}{k} \left(\frac{\partial g_M}{\partial \theta_k} + iMg_M \right) \right] - 2i \sum_{M'=-\infty}^{\infty} e\mathbf{E}_{dc} \cdot (g_{M-M'} \mathbf{X}_{M'} - f_{M-M'} \mathbf{Y}_{M'}).
\end{aligned} \tag{D9}$$

The phenomenological relaxation rates $\tau_{F,M}^{-1}$ and $\tau_{f,M}^{-1}$ regularize the longitudinal responses and describe the scattering required for a steady state under a DC drive. Interband coherence determines the orbital Hall responses, which remain well-defined in the ideal clean limit.

For linear response to E_{dc} , the zeroth-order density matrix is diagonal in the periodic Floquet-mode basis Eq. (D1):

$$F_M^{(0)} = \frac{\delta_{M0}}{2}(n_+ + n_-), \quad f_M^{(0)} = \frac{\delta_{M0}}{2}(n_+ - n_-), \quad g_M^{(0)} = 0, \tag{D10}$$

where $n_{\pm}$ denote the occupations assigned to the two Floquet modes. The frozen zero-temperature occupations are $n_{\pm} = f(\varepsilon_{\pm}(k))$. Thus $n_- - n_+$ is given by Eq. (8). This approximation neglects drive-induced heating and bath-dependent redistribution. To first order in E_{dc} , the density-matrix components are

$$\begin{aligned}
F_M &= \frac{\delta_{M0}}{2}(n_+ + n_-) - \frac{\delta_{M0}}{2} \tau_{F,0} eE_{dc} \cos \theta_k \cdot \partial_k (n_+ + n_-) + O(E_{dc}^2), \\
f_M &= \frac{\delta_{M0}}{2}(n_+ - n_-) - \frac{\delta_{M0}}{2} \tau_{f,0} eE_{dc} \cos \theta_k \cdot \partial_k (n_+ - n_-) + O(E_{dc}^2), \\
g_M &= \frac{e\mathbf{E}_{dc} \cdot \mathbf{Y}_M}{\Delta - M\Omega_0} (n_+ - n_-) + O(E_{dc}^2).
\end{aligned} \tag{D11}$$

Substituting Eq. (D7) and transforming back to the band basis by

$$\rho^{\text{band}}(\mathbf{k}, t) = \mathbf{V}(\mathbf{k}, t) \rho^F(\mathbf{k}, t) \mathbf{V}^\dagger(\mathbf{k}, t), \tag{D12}$$

and averaging over one driving period $T = 2\pi/\Omega_0$ yields

$$\begin{aligned}
\text{Re}\langle \rho_{+-}^{\text{band}}(\mathbf{k}) \rangle_{\text{cycle}} &= \frac{keA_0}{m_2 \Omega_0} \sin \phi (n_- - n_+) - \cos \theta_k eE_{dc} \frac{keA_0 \sin \phi}{m_2 \Omega_0} \tau_{f,0} \partial_k (n_- - n_+) \\
&+ \sin \theta_k eE_{dc} \frac{(n_- - n_+)}{k} \left(\frac{\Omega_0 \cos \phi - \Delta(1 + \cos^2 \phi)/2}{\Omega_0^2 - \Delta^2} - \frac{keA_0 \sin \phi \cos \phi}{m_2 \Omega_0 \Delta} \right).
\end{aligned} \tag{D13}$$

Appendix E: Evaluation of the conductivities

For the longitudinal orbital conductivity, the derivative of the occupation difference reduces the radial integral to the two Fermi contours:

$$\partial_k(n_+ - n_-) = \delta(k - k_{F-}) - \delta(k - k_{F+}), \quad (\text{E1})$$

which gives

$$\begin{aligned} \sigma_{L_z}^L &= \frac{J_{L_z}^L}{E_{\text{dc}}} = \frac{e^2 \tau_{f,0} A_0}{\pi m_1 m_2 \Omega_0} (k_{F-}^3 \sin \phi(k_{F-}) - k_{F+}^3 \sin \phi(k_{F+})) \\ &= -\frac{e^3 \tau_{f,0} A_0^2}{\pi m_1 m_2 \Omega_0} \left(\frac{k_{F-}^4}{\sqrt{(k_{F-} e A_0)^2 + (k_0^2 - k_{F-}^2)^2}} + \frac{k_{F+}^4}{\sqrt{(k_{F+} e A_0)^2 + (k_0^2 - k_{F+}^2)^2}} \right) \\ &= -\frac{e^3 \tau_{f,0} A_0^2}{\pi m_1 m_2 \Omega_0} (G_L(k_{F-}) + G_L(k_{F+})), \quad G_L(k) = \frac{k^4}{\sqrt{(k_0^2 - k^2)^2 + (e A_0 k)^2}}. \end{aligned} \quad (\text{E2})$$

For the orbital Hall conductivity, the two terms in the last two lines of Eq. (41) can be simplified as

$$\frac{\Omega_0 \cos \phi - \Delta(1 + \cos^2 \phi)/2}{\Omega_0^2 - \Delta^2} = \frac{m_2}{2} \frac{k_0^2 \cdot \frac{k_0^2 - k^2}{\sqrt{(k_0^2 - k^2)^2 + (k e A_0)^2}} + \frac{\sqrt{(k_0^2 - k^2)^2 + (k e A_0)^2}}{2}}{k_0^4 - (k_0^2 - k^2)^2 - (k e A_0)^2} \left(1 + \frac{(k_0^2 - k^2)^2}{(k_0^2 - k^2)^2 + (k e A_0)^2} \right) \text{sgn}(k_0 - k), \quad (\text{E3})$$

and

$$\frac{\sin \phi \cos \phi}{\Delta} = \frac{m_2}{2} \frac{k e A_0 \cdot (k_0^2 - k^2)}{((k e A_0)^2 + (k_0^2 - k^2)^2)^{3/2}} \text{sgn}(k_0 - k), \quad (\text{E4})$$

Combining these identities and splitting the radial integral at k_0 , as required by $k_{F+} < k_0 < k_{F-}$, gives

$$\sigma_{L_z}^H = \frac{J_{L_z}^H}{E_{\text{dc}}} = \frac{m_2 e}{2\pi m_1} \left[\int_{k_{F+}}^{k_0} \frac{dG_H}{dk} dk - \int_{k_0}^{k_{F-}} \frac{dG_H}{dk} dk \right] = \frac{m_2 e}{2\pi m_1} [2G_H(k_0) - G_H(k_{F-}) - G_H(k_{F+})]. \quad (\text{E5})$$

where

$$G_H(k) = \int \left[\frac{k_0^2(k_0^2 - k^2) + (k_0^2 - k^2)^2 + \frac{1}{2}(k e A_0)^2}{\sqrt{(k_0^2 - k^2)^2 + (k e A_0)^2} (k_0^4 - (k_0^2 - k^2)^2 - (k e A_0)^2)} - \frac{(e A_0)^2}{2k_0^2} \frac{k^2(k_0^2 - k^2)}{((k_0^2 - k^2)^2 + (k e A_0)^2)^{3/2}} \right] \cdot k dk. \quad (\text{E6})$$

Explicit integration gives

$$\begin{aligned} G_H(k) &= \frac{2k_0^2}{2k_0^2 - (e A_0)^2} \tanh^{-1} \frac{k^2 - \sqrt{(k^2 - k_0^2)^2 + (e A_0 k)^2}}{k_0^2} \\ &\quad + \frac{(e A_0)^4}{2k_0^2(2k_0^2 - (e A_0)^2)} \tanh^{-1} \frac{2k_0^2 - k^2 - (e A_0)^2 + \sqrt{(k^2 - k_0^2)^2 + (e A_0 k)^2}}{k_0^2} \\ &\quad - \frac{2k_0^2 - (e A_0)^2}{4k_0^2} \ln \left[2\sqrt{(k^2 - k_0^2)^2 + (e A_0 k)^2} + 2(k^2 - k_0^2) + (e A_0)^2 \right] \\ &\quad + \frac{(e A_0)^2}{2k_0^2} \frac{k_0^4 - 3(k k_0)^2 + (k e A_0)^2}{(4k_0^2 - (e A_0)^2) \sqrt{(k^2 - k_0^2)^2 + (e A_0 k)^2}}. \end{aligned} \quad (\text{E7})$$

Therefore

$$\begin{aligned}
G_H(k_0) &= \frac{2k_0^2}{2k_0^2 - (eA_0)^2} \tanh^{-1} \left(1 - \frac{eA_0}{k_0} \right) - \frac{2k_0^2 - (eA_0)^2}{4k_0^2} \ln(2eA_0k_0 + (eA_0)^2) + \frac{eA_0((eA_0)^2 - 2k_0^2)}{2k_0(4k_0^2 - (eA_0)^2)} + O(A_0^4) \\
&= -\ln(eA_0) - \frac{3eA_0}{4k_0} + O(A_0^2), \\
G_H(k_{F+}) &= \frac{1}{2} \ln \frac{k_{F+}^2}{k_0^2 - k_{F+}^2} - \frac{1}{2} \ln \frac{(eA_0)^2 k_0^2}{k_0^2 - k_{F+}^2} + O(A_0^2) \\
&= \ln \frac{k_{F+}}{eA_0 k_0} + O(A_0^2), \\
G_H(k_{F-}) &= \frac{1}{2} \ln \frac{4k_0^2(k_{F-}^2 - k_0^2)}{(eA_0)^2 k_{F-}^2} - \frac{1}{2} \ln [4(k_{F-}^2 - k_0^2)] + O(A_0^2) \\
&= \ln \frac{k_0}{eA_0 k_{F-}} + O(A_0^2),
\end{aligned} \tag{E8}$$

and

$$2G_H(k_0) - G_H(k_{F+}) - G_H(k_{F-}) = \ln \frac{k_{F-}}{k_{F+}} - \frac{3eA_0}{2k_0} + O(A_0^2). \tag{E9}$$

Substituting into Eq. (E5) gives Eq. (44).